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Lecture 27 Principal Component Analysis I

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

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2. Principal Components Analysis
3. Singular Value Decomposition
4. Summary and Remark

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Introduction

- Principal Component Analysis (PCA) produces a low-dimensional representation of a dataset. It finds a sequence of linear combinations of the variables that have maximal variance, and are mutually uncorrelated.
- Apart from producing derived variables for use in supervised learning problems, say dimension reduction, PCA also serves as a tool for data visualization.
- PCA was invented in 1901 by Karl Pearson, as an analogue of the principal axis theorem in mechanics; it was later independently developed (and named) by Harold Hotelling in the 1930s.
- other names: Kosambi-Karhunen-LoÃ¨ve transform (KLT), spectral decomposition ...

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Principal Components Analysis

- The first principal component of a set of features $X_1, \cdots, X_p$ is the normalized linear combination of the features

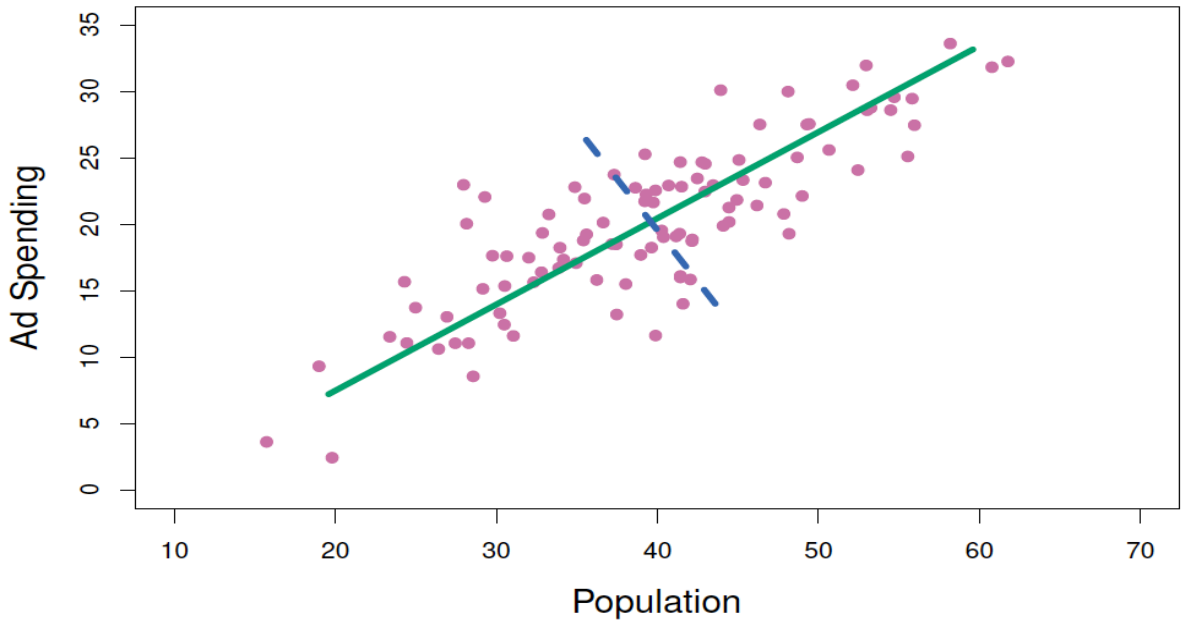
$$Z_1 = \phi_{11} X_1 + \cdots + \phi_{p1} X_p$$

that has the largest variance. By normalized, we mean that $\sum_{j=1}^{p} \phi_{j1}^2 = 1$.

- We refer to the elements $\phi_{11}, \cdots, \phi_{p1}$ as the loadings of the first principal component; together, the loadings make up the principal component loading vector, $\phi_1 = (\phi_{11}, \cdots, \phi_{p1})^T$.
- We constrain the loadings so that their sum of squares is equal to one, since otherwise setting these elements to be arbitrarily large in absolute value could result in an arbitrarily large variance.

An Example

- The population size (pop) and ad spending (ad) for 100 different cities are shown as purple circles.
- The green solid line indicates the first principal component direction, and the blue dashed line indicates the second principal component direction.



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Computation of Principal Components

- Suppose we have a $n \times p$ data set $\boldsymbol{X}$. Since we are only interested in variance, we assume that each of the variables in $\boldsymbol{X}$ has been centered to have mean zero (that is, the column means of $\boldsymbol{X}$ are zero).
- We then look for the linear combination of the sample feature values of the form

$$z_{i1} = \phi_{11} x_{i1} + \cdots + \phi_{p1} x_{ip}, \tag{2.1}$$

  for $i = 1, \cdots, n$ that has largest sample variance, subject to the constraint that $\sum_{j=1}^{p} \phi_{j1}^2 = 1$.
- Since each of the x_{ij} has mean zero, then so does z_{i1} (for any values of ϕ_{j1}). Hence the sample variance of the z_{i1} can be written as $1/n \sum_{i=1}^{n} z_{i1}^2$.

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Computation of Principal Components

- Plugging in (2.1) the first principal component loading vector solves the optimization problem

$$\text{maximize}_{\phi_{11},\cdots,\phi_{p1}} 1/n \sum_{i=1}^{n} \left(\sum_{j=1}^{p} \phi_{j1} x_{ij} \right)^2, \quad \text{subject to} \sum_{j=1}^{p} \phi_{j1}^2 = 1. \tag{2.2}$$

- This problem can be solved via a singular-value decomposition of the matrix $\boldsymbol{X}$, a standard technique in linear algebra.
- We refer to $\boldsymbol{Z}_1$ as the first principal component, with realized values $z_{11}, \cdots, z_{n1}$.

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Geometry of PCA

- The loading vector ϕ_1 with elements $\phi_{11}, \cdots, \phi_{p1}$ defines a direction in feature space along which the data vary the most.
- If we project the n data points $x_1, \cdots, x_n$ onto this direction, the projected values are the principal component scores $z_{11}, \cdots, z_{n1}$ themselves.

Further principal components

- The second principal component is the linear combination of $X_1, \cdots, X_p$ that has maximal variance among all linear combinations that are uncorrelated with Z_1.
- The second principal component scores $z_{12}, \cdots, z_{n2}$ take the form

$$z_{i2} = \phi_{12} x_{i1} + \cdots + \phi_{p2} x_{ip},$$

where ϕ_2 is is the second principal component loading vector, with elements $\phi_{12}, \cdots, \phi_{p2}$.

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Further principal components

- It turns out that constraining Z_2 to be uncorrelated with Z_1 is equivalent to constraining the direction ϕ_2 to be orthogonal (perpendicular) to the direction ϕ_1. And so on.
- The principal component directions $\phi_1, \phi_2, \cdots$ are the ordered sequence of right singular vectors of the matrix $\boldsymbol{X}$, and the variances of the components are $1/n$ times the squares of the singular values.
- There are at most $\min(n-1, p)$ principal components.

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Singular Value Decomposition

- In linear algebra, the singular value decomposition (SVD) is a factorization of a real or complex matrix.
- In statistics, given data matrix $\boldsymbol{X}$ with dimension $n \times p$, its is a factorization of the form
$$\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{V}^T,$$
where $\boldsymbol{U}$ is $n \times n$ unitary matrix ($\boldsymbol{U}\boldsymbol{U}^T = \boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}$), $\boldsymbol{\Lambda}$ is $n \times p$ rectangular diagonal matrix with non-negative real numbers on the diagonal, and $\boldsymbol{V}$ is $p \times p$ unitary matrix.
- The diagonal entries λ_i of $\boldsymbol{\Lambda}$ are known as the singular values of $\boldsymbol{X}$. The columns of $\boldsymbol{U}$ and the columns of $\boldsymbol{V}$ are called the left singular vectors and right singular vectors of $\boldsymbol{X}$, respectively.
- We assume that $n > p$ and $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$.

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Singular Value Decomposition

- The left-singular vectors of $\boldsymbol{X}$ are a set of orthonormal eigenvectors of $\boldsymbol{X}\boldsymbol{X}^T$.
- The right-singular vectors of $\boldsymbol{X}$ are a set of orthonormal eigenvectors of $\boldsymbol{X}^T\boldsymbol{X}$.
- The non-zero singular values of $\boldsymbol{X}$ (found on the diagonal entries of $\boldsymbol{\Lambda}$) are the square roots of the non-zero eigenvalues of both $\boldsymbol{X}^T\boldsymbol{X}$ and $\boldsymbol{X}\boldsymbol{X}^T$.
- We have
$$\boldsymbol{X}^T\boldsymbol{X} = \boldsymbol{V}\boldsymbol{\Lambda}^2\boldsymbol{V}^T, \text{ and } \boldsymbol{X}\boldsymbol{X}^T = \boldsymbol{U}\boldsymbol{\Lambda}^2\boldsymbol{U}^T.$$

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SVD and PCA

- The optimization (2.2) is equivalent to

$$\text{maximize}_{\|\phi\|=1} \left\{ \|\boldsymbol{X}\phi\|^2 \right\} = \text{maximize}_{\|\phi\|=1} \left\{ \phi^T \boldsymbol{X}^T \boldsymbol{X} \phi \right\}.$$

- This can be rewritten as

$$\text{maximize}_{\|\phi\|=1} \left\{ \phi^T \boldsymbol{V} \boldsymbol{\Lambda}^2 \boldsymbol{V}^T \phi \right\},$$

- Let $\psi = \boldsymbol{V}^T \phi$, then $\|\phi\| = 1$ is equivalent to $\|\psi\| = 1$ and the above can be simplified as

$$\text{maximize}_{\|\psi\|=1} \left\{ \psi^T \boldsymbol{\Lambda}^2 \psi \right\} = \text{maximize}_{\|\psi\|=1} \sum_{j=1}^{p} \lambda_j^2 \psi_j^2 = \lambda_1^2,$$

where we assume $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$. The equality holds when $\psi = (1, 0, \cdots, 0)^T$ that is $\phi = v_1$, the first column of the right-singular vectors $\boldsymbol{V}$.

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SVD and PCA

- The second PCA can be calculated in a similar way and it is the second column of $\boldsymbol{V}$ corresponding to the second largest singular value.
- The rest PCA can be done accordingly. Therefore $\phi_j = v_j$.
- Due to the properties of SVD, all the PCAs are orthogonal, that is $v_j^T v_{j'} = 1$ if $j = j'$ and 0 otherwise.

Summary and Remark

- PCA
- SVD
- Read textbook Chapter 10 and code